CSCI 3052U • MACHINE LEARNING I

Out-of-Distribution Generalization in Satellite Land Cover Classification

Team SixtyFive.AI

Project Overview

Project Overview and Dataset

- **Objective**
  - Investigate how geographic differences affect satellite image classification.
- **Dataset**
  - EuroSAT Sentinel-2 satellite imagery.
- **Samples**
  - 27,000 images with RGB and multispectral representations.
- **Classes**
  - 10 land-cover categories, including Forest, Residential, River, and Annual Crop.
- **Task**
  - Supervised multiclass classification.
- **Goal**
  - Compare development validation performance against geographically separated OOD performance.

DATASET VISUALIZATION

Representative RGB Samples



Methodology & Pipeline

Current Methods and Approaches

01

Data Partitioning

Geographic metadata was used to create provisional development and OOD partitions.

02

Baseline Setup

A majority-class classifier was implemented as a simple baseline.

03

Image Preprocessing

RGB images were resized to 16 × 16 pixels.

04

Dimensionality Reduction

Principal Component Analysis (PCA) reduced the input to 100 components.

05

Model Training

A Random Forest classifier with 200 decision trees was trained.

06

Validation Strategy

Development training and validation used an 80/20 stratified split with random seed 42.

DATASET COVERAGE

Geographic Distribution across Europe



01

Sentinel-2 Satellite Imagery

Multispectral land cover samples gathered across 34 European nations.

02

Spatial & Climatic Diversity

Captures wide topographical variance, seasonal nuances, and distinct biomes.

03

OOD Evaluation Basis

Provides regional partitions to benchmark out-of-distribution generalization capability.

Theoretical Framework

Theoretical Justification for Our Methods

01

PCA

Reduces dimensionality by retaining directions of high variance.

02

Decision Trees

Learn decision rules that divide the feature space into classes.

03

Random Forest

Combines multiple decision trees to improve prediction stability and reduce overfitting.

04

Macro-F1 Metric

Measures classification performance while giving equal importance to all 10 classes.

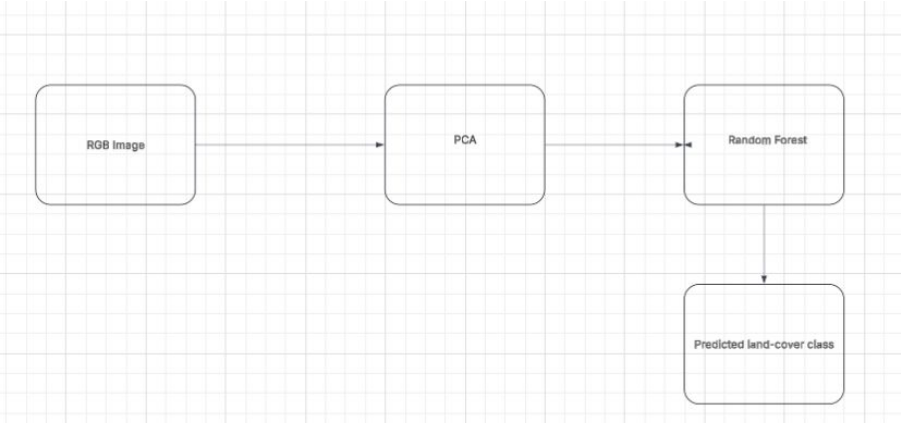
05

ResNet-18 (Planned)

A convolutional neural network intended to learn more complex spatial features from images.

PIPELINE ARCHITECTURE

Baseline Machine Learning Pipeline



01

RGB Image Input

Multi-channel satellite imagery input for classification.

02

PCA Dimensionality Reduction

Extracts key spectral features while reducing noise.

03

Random Forest Classifier

Ensemble decision trees predict land cover categories.

04

Predicted Land-Cover Class

Outputs final thematic ground coverage labels.

Preliminary Findings

Preliminary Results & Performance Analysis

Dev Macro-F1

0.6910

PCA + Random Forest

OOD Test Set

0.4365

Geographic Out-of-Distribution

Performance Drop

-0.2546

Macro-F1 Decrease

01

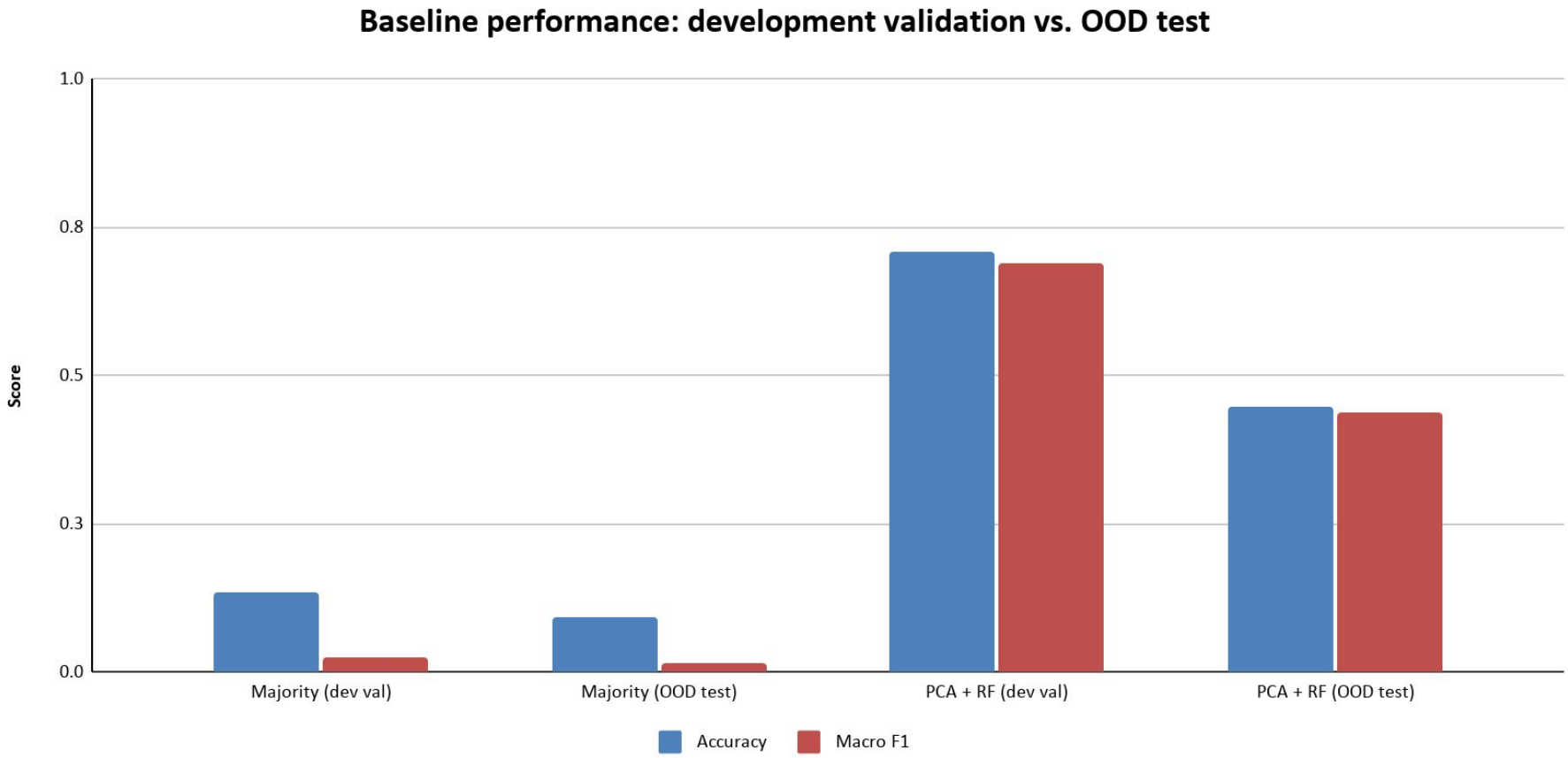
Reduced Generalization

Results suggest significantly reduced model generalization when evaluated on geographically distinct satellite images.

02

Geographic Split Validation

These findings are preliminary and explicitly require further rigorous geographic split validation to establish robustness.


Baseline performance: development validation vs. OOD test
Score
1.0
0.8
0.5
0.3
0.0
Majority (dev val)
Majority (OOD test)
PCA + RF (dev val)
PCA + RF (OOD test)
Accuracy
Macro F1

Project Risks & Constraints

Current Challenges and Deviations

01

Geographic Boundaries

Some image locations remain ambiguous or require additional verification.

02

Spatial Leakage

Nearby image samples could cause overly optimistic evaluation results.

03

Class Differences

Geographic regions may contain different distributions of land-cover classes.

04

Validation Strategy

The current baseline uses a stratified development split; geographic-grouped validation is still planned.

05

Model Development

ResNet-18 experiments remain part of the next phase.

Project Schedule

Timeline & Next Steps

OCT 19

Baseline Experiments

Baseline experiments and documentation completed.

Completed

OCT 30

Midpoint Review

Midpoint presentation and progress review.

Milestone

NOV 20

ResNet-18 & Validation

Complete ResNet-18 experiments, geographic validation, and class-level error analysis.

In Progress

DEC 04

Final Deliverable

Submit final project report and demonstration.

Upcoming

Team Progress and Conclusion

Midpoint Check-in & Project Summary

01

Midpoint Peer Check-in

- **Mohammad Jafari:** EDA, geographic prep, docs, repo integration
- **Rayan Khan:** EDA, geographic prep, docs, repo integration
- **Haseeb Sajjad:** baseline model development & docs
- **Faraz Shaikh:** baseline model development & docs
- **Chris Ibrahim:** baseline model development & docs

02

Conclusion & Highlights

Dataset & Baselines

Completed dataset preparation and baseline experiments.

Performance Insight

Observed a substantial reduction in geographic OOD performance.

Next Focus

Improve evaluation reliability and test planned ResNet-18 model.

Peer Review Feedback (Team 27)

Received feedback recommending strengthened geographic validation and class-level error analysis. Both will be addressed in Milestone 5.